

Synthesis and properties of multiprincipal component AlCoCrFeNiSi_x alloys

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ABSTRACT

By introducing nonmetallic Si element, AlCoCrFeNiSi_x (x values in molar ratio, $x=0, 0.2, 0.4, 0.6, 0.8$ and 1.0) alloys are designed and prepared. One kind of nanoscale cellular structure forms due to the introduction of nonmetallic Si element. The alloy is composed mainly of BCC solid solution and possesses excellent room-temperature compressive mechanical properties. The excessive introduction of Si element leads to the precipitation of δ phase at grain boundary. The improvement of mechanical properties is attributed to the solid solution of Si element and precipitation strengthening of nanoscale cellular structure. The precipitated δ phase at grain boundary should be responsible for transition of alloy from plastic fracture to brittle fracture. Among them, Si_{0.4} alloy exhibits excellent comprehensive mechanical properties, yield stress of 1481 MPa, fracture strength of 2444 MPa and plastic strain of 13.38%, respectively.

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1. Introduction

In recent years, one kind of new alloy, high-entropy alloy with multiple principal elements in equimolar or near-equimolar ratios, has received more and more attentions due to its unique structure and excellent properties [1–7]. Promising properties in hardness, wear resistance, oxidation resistance and corrosion resistance can be obtained through proper composition design [4,8–10]. These discoveries bring new ideas to material design and stimulate a new conception of material research. A series of multiprincipal component alloy systems have been explored in the past decade [11–14]. The most commonly used elements in alloy systems that have been investigated are Al, Ti, Cr, Fe, Co, Ni and Cu. These elements except Al are transition metal elements of the fourth period. It is known that the proper addition of elements like C and Si with small atomic radii in traditional materials like steel is beneficial to their structure and properties [15–18]. However, the effects of nonmetallic elements on the multiprincipal component alloys are rarely investigated and reported. Therefore, in this paper, the effects of Si element on the microstructure and properties of AlCoCrFeNi multiprincipal component alloy are investigated systematically. The introduction of Si element changes the alloy structure evidently, which further leads to the improvement of alloy properties. The structure evolution and strengthening mechanism of alloy are discussed in detail.

2. Experimental

Alloy ingots with nominal compositions of AlCoCrFeNiSi_x (x values in molar ratio, $x=0, 0.2, 0.4, 0.6, 0.8$ and 1.0 , denoted by Si₀, Si_{0.2}, Si_{0.4}, Si_{0.6}, Si_{0.8} and Si_{1.0}, respectively) were prepared by arc melting a mixture of ultrasonically cleansed Al, Co, Cr, Fe, Ni and Si elements with a purity of above 99.9 wt% in a water-cooled copper hearth under Ti-gettered high purity argon atmosphere. The chemical homogeneity was realized by repeated melting at least four times. The proper amount of ingots were then remelted under high vacuum in a quartz tube by using induction heating coil and injected through a nozzle with 0.5–1 mm in diameter into the copper mould with a cavity of 5 mm in diameter and 50 mm in height. The phases and microstructures were characterized using X-ray diffractometer (XRD, Philips PW1050, Cu K α), scanning electron microscope (SEM, Hitachi S3400N) accompanied with energy dispersive spectrometer and transmission electron microscope (TEM, JEOL 2010, 200 kV). The end surfaces of compressive samples with 5 mm in diameter and 8 mm in length were polished on 2000 grit sand papers. The microstructure of alloy was revealed by etching in aqua regia. Thin slices from as-cast rods were used for preparing the TEM samples, which were ground and mechanically dimpled with a GATAN precision dimple grinder as well as use argon ion milling as the final thinning process by a GATAN precision ion polishing system (PIPS). Compression test was performed on MTS 800 machine with a strain rate of $1 \times 10^{-4} \text{ s}^{-1}$.

3. Results

Fig. 1 shows the XRD patterns of as-cast AlCoCrFeNiSi_x alloys. It is evident that all alloys are composed mainly of BCC solid solu-

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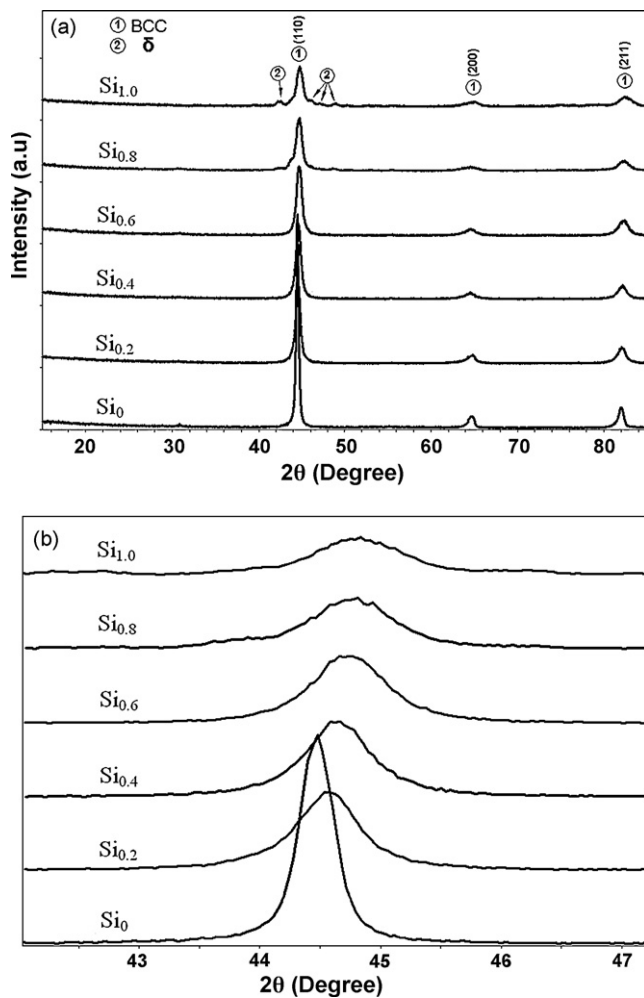


Fig. 1. XRD patterns of as-cast AlCoCrFeNiSi_x ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1.0) alloys.

tion. The XRD patterns of Si₀, Si_{0.2}, Si_{0.4} and Si_{0.6} exhibit only single BCC phase reflection. For Si_{0.8} and Si_{1.0} alloys, some tiny peaks of δ phases appear on the curves. As Si content increases from 0.8 to 1.0, the relative intensity of δ phase becomes stronger. Fig. 1(b) is an enlarged picture of the main peak. The peaks of BCC phase gradually shift rightwards with the increase of Si content, which indicates that the lattice parameter of BCC phase decreases slightly. It can be deduced that Si element with small atomic radius exists mainly in solid solution by substituting other elements.

Fig. 2 shows the back-scattering images of as-cast AlCoCrFeNiSi_x alloys. Si₀ and Si_{0.2} alloys exhibit single even featureless solid-solution structure, which is consistent with the XRD result. When Si content is 0.4 and 0.6, two distinct regions (A and B in Fig. 2(c and d)) in alloy form due to constitutional segregation. The EDX results in Table 1 show that the composition of region A is similar to the nominal composition of alloy. The Al content of region B is higher than the nominal composition of alloy, the Cr and Si contents of region B are lower than the nominal composition of alloy. When Si content is 0.8 and 1.0, the grain boundary becomes clear and coarse. The EDX results show that the grain boundary is rich in Cr element and poor in Al element.

Fig. 3 shows the TEM bright-field images with corresponding selected area electron diffraction (SAED) patterns of Si₀, Si_{0.2}, Si_{0.6}, Si_{0.8} and Si_{1.0} alloys. It is found that Si₀ alloy exhibits a single BCC solid-solution structure, as shown in Fig. 3(a). It is worth noting that a uniform nanoscale cellular structure is precipitated in the grain of Si_{0.2} alloy through spinodal decomposition. The width and

length of precipitated phase are about 45–75 nm and 90–230 nm, respectively. The spacing between particles is about 10 nm, which is also observed in Si_{0.6}, Si_{0.8} and Si_{1.0} alloys. On the other hand, δ phase is observed in the grain boundary of Si_{0.6}, Si_{0.8} and Si_{1.0} alloys, as shown in Fig. 3(d, f and h). Its size increases as Si content increases from 0.6 to 1.0.

Fig. 4 shows the room-temperature compressive true stress–strain curves of as-cast AlCoCrFeNiSi_x alloys. It is found that the alloy strength increases obviously as Si content increases. However, the alloy ductility is weakened at the same time. For Si_{0.2} and Si_{0.4} alloys, they both exhibit good yield stress, fracture strength, plastic deformation and obvious work hardening behavior. The alloy strength continues to be improved evidently accompanying the sacrifice of ductility as Si content increases from 0.6 to 1.0. The yield strength σ_y , fracture strength σ_{max} and plastic strain limit ε_p of alloys are listed in Table 2. Among them, Si_{0.4} alloy exhibits excellent comprehensive mechanical properties, including yield stress of 1481 MPa, fracture strength of 2444 MPa and plastic strain of 13.38%, respectively. For Si_{1.0} alloy, the yield stress and fracture strength are as high as 2411 MPa and 2950 MPa respectively.

4. Discussion

According to the Gibbs phase rule, $F = C - P + 1$ (F : degree of freedom, C : number of components, and P : number of phase), the maximum number of equilibrium phase in a C component system at constant pressure is $P = C + 1$ [19]. However, in this research, simple solid-solution phase such as BCC are prior to intermetallic phase to form in AlCoCrFeNiSi_x alloys, and the total number of phases is well below the maximum equilibrium number allowed by the Gibbs phase rule. The prior formation of simple solid solution to intermetallic compound in this alloy system can be attributed to the effect of high entropy of mixing [6]. The high mixing entropy enhances the mutual elemental solubility and prevents phase separation into terminal phases. The mixing enthalpy values of Si–Al, Si–Co, Si–Cr, Si–Fe and Si–Ni pairs are -2 , -21 , -20 , -18 and -23 kJ/mol, respectively [20]. It is clear that they are large negative values. Therefore, it can be deduced that Si element has strong bonding with other elements. Some elements will be replaced by Si element. The number of atoms replaced by Si will increase as Si content increases. From the above analysis, the Al and Cr elements are mainly substituted by Si element, and segregate in grain and grain boundary.

The improved properties of alloys are attributed to solid solution of Si element and precipitation strengthening of nanoscale cellular structure. For cubic crystals, the lattice strain can be expressed as $\varepsilon = \Delta a/a_0$, where $\Delta a = |a - a_0|$, a and a_0 is the lattice constant of actual alloy crystal and nondeformed perfect crystal, respectively [21]. Here, the crystal of Si₀ alloy is regarded as the perfect crystal to investigate the effect of Si addition on the change of ε . Table 3 lists the calculated lattice constants of AlCoCrFeNiSi_x alloys base on the XRD results. It is found that the lattice constants of BCC phase decrease as Si content increases from 0 to 1.0, which results in the increase of ε . Therefore, it is deduced that the lattice distortion energy will increase significantly and the effect of solid-solution strengthening is enhanced. Thus, the alloy strength increases greatly, with the decrease of ductility. The TEM images show nano-level cellular precipitated phase forms in as-cast alloy. The formation process of nanoscale cellular structure is described as follow: liquid phase transforms into solid-solution phase through solidification with segregation. And then solid-solution phase appears with spinodal decomposition to form nanospaced modulated structure. Precipitation in high-entropy alloy involves partitioning of a lot of elements and substitutional diffusion. Nucleation and growth of phase are delayed, which is beneficial to the formation of nanoscale cellular structure. The for-

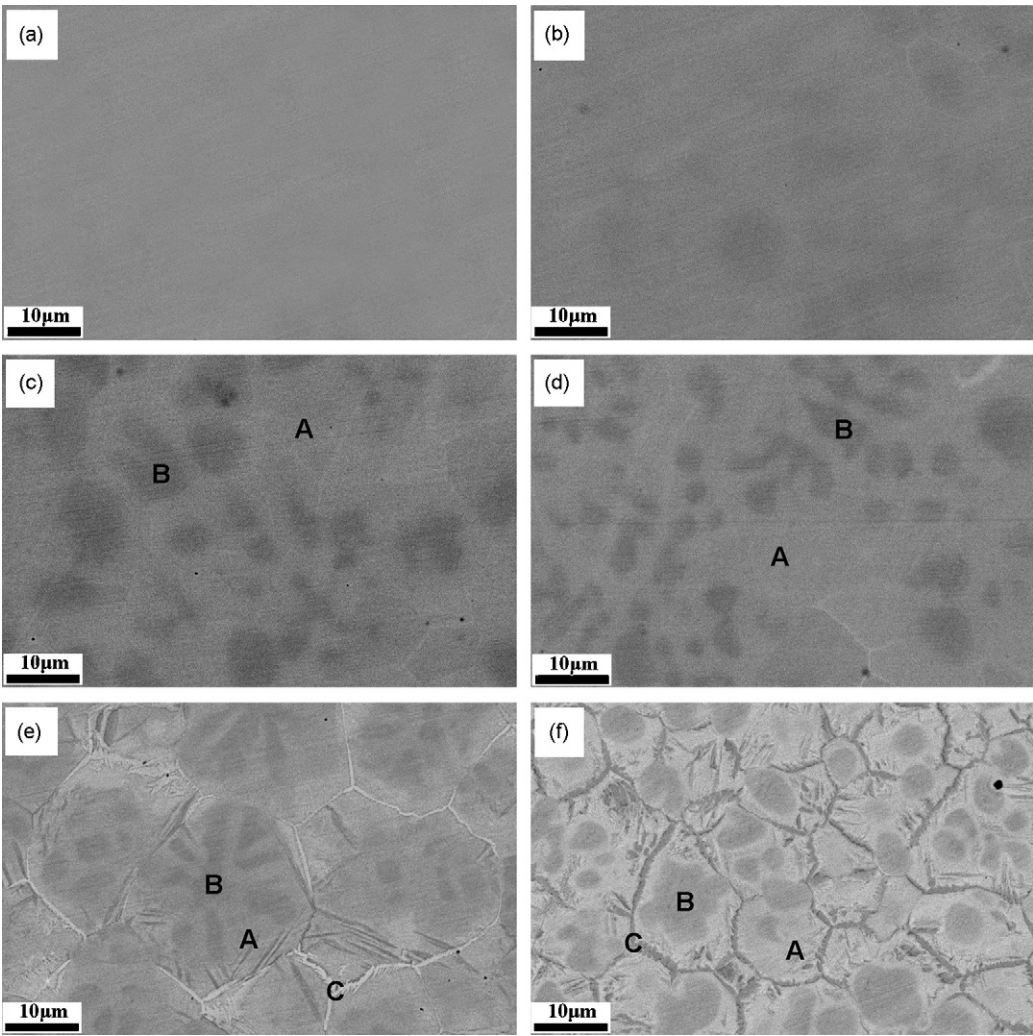


Fig. 2. Back-scattering images of as-cast AlCoCrFeNiSi_x alloys: (a) 0, (b) 0.2, (c) 0.4, (d) 0.6, (e) 0.8 and (f) 1.0.

Table 1
Compositions (at.%) of areas in as-cast AlCoCrFeNiSi_x ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1.0) alloys and atomic radii of different elements.

x	Regions in Fig. 2	Al	Co	Cr	Fe	Ni	Si
		1.43 Å	1.25 Å	1.21 Å	1.24 Å	1.24 Å	1.17 Å
0	Nominal	20.0	20.0	20.0	20.0	20.0	–
	EDX	16.7	21.1	21.2	21.9	19.1	–
0.2	Nominal	19.2	19.2	19.2	19.2	19.2	3.8
	EDX	17.6	20.1	20.2	19.9	18.5	3.7
0.4	Nominal	18.5	18.5	18.5	18.5	18.5	7.4
	A	11.5	17.2	25.4	22.9	14.1	8.9
	B	29.9	20.2	9.6	12.5	25.3	2.5
0.6	Nominal	17.8	17.8	17.8	17.8	17.8	10.7
	A	10.5	17.2	24.4	21.0	13.1	13.8
	B	31.8	19.2	8.6	11.5	24.3	4.6
0.8	Nominal	17.2	17.2	17.2	17.2	17.2	13.8
	A	13.5	19.7	20.1	19.7	14.0	16.1
	B	32.2	18.7	7.6	11.0	23.9	6.6
	C	4.0	15.0	30.2	18.9	10.9	20.8
1.0	Nominal	16.6	16.6	16.6	16.6	16.6	16.6
	A	17.3	18.3	11.1	19.8	16.9	17.3
	B	32.8	18.5	6.7	10.4	23.4	8.2
	C	2.4	5.6	53.8	10.9	3.7	23.6

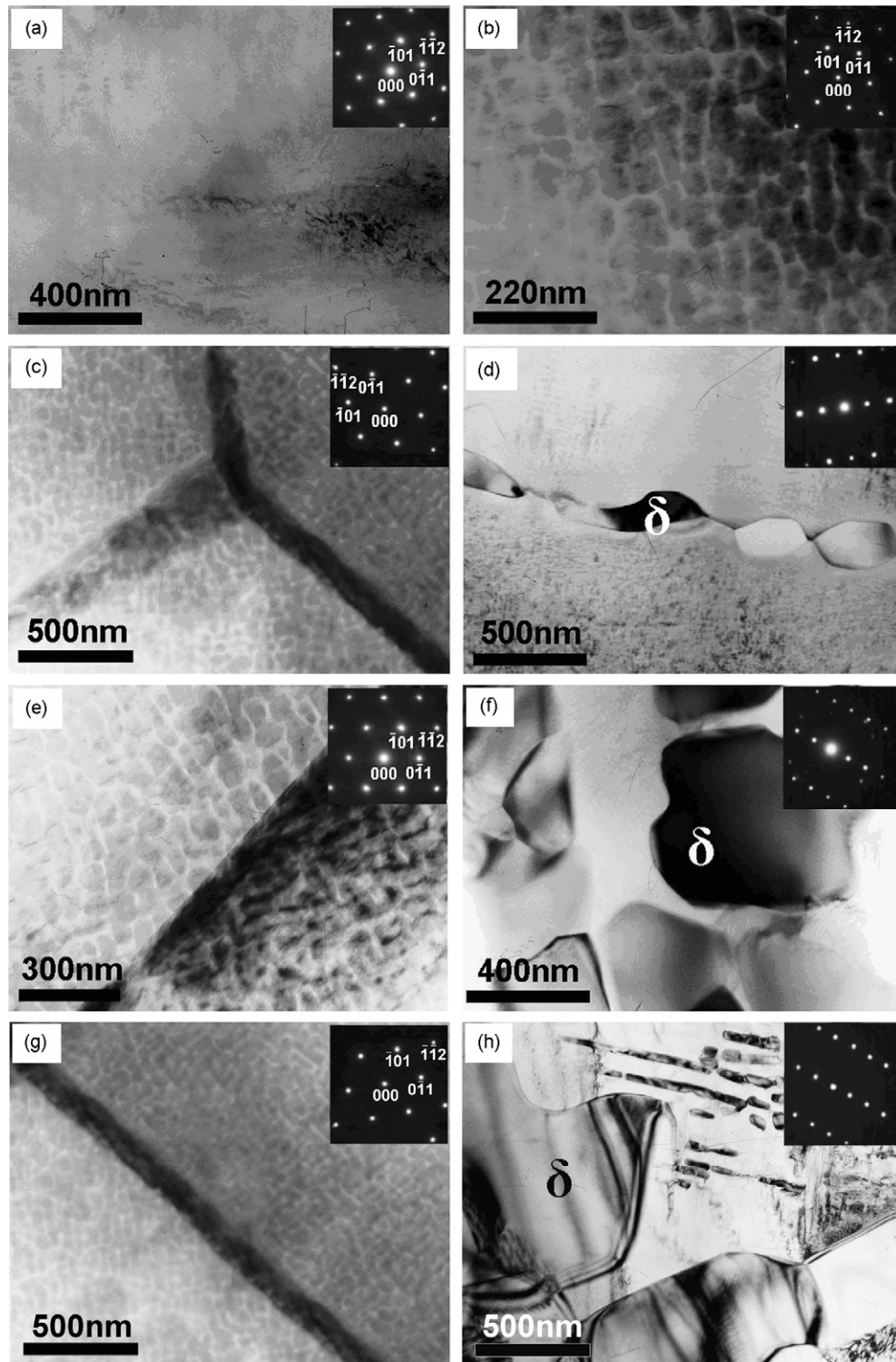


Fig. 3. TEM images with corresponding SAED patterns (inserted pictures) of as-cast AlCoCrFeNiSi_x alloys: (a) 0, (b) 0.2, (c) 0.6, (d) 0.6 (grain boundary and SAED pattern of δ phase), (e) 0.8, (f) 0.8 (C in Fig. 2(e) and SAED pattern of δ phase) (g) 1.0, (h) 1.0 (C in Fig. 2(f) and SAED pattern of δ phase).

Table 2

Mechanical properties of as-cast AlCoCrFeNiSi_x ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1.0) alloys.

x	Yield stress σ_y (MPa)	Compressive strength $\sigma_{\max}$ (MPa)	Plastic strain ε_p (%)
0	1110	–	–
0.2	1265	2173	13.76
0.4	1481	2444	13.38
0.6	1834	2195	2.56
0.8	2179	2664	1.77
1.0	2411	2950	1.17

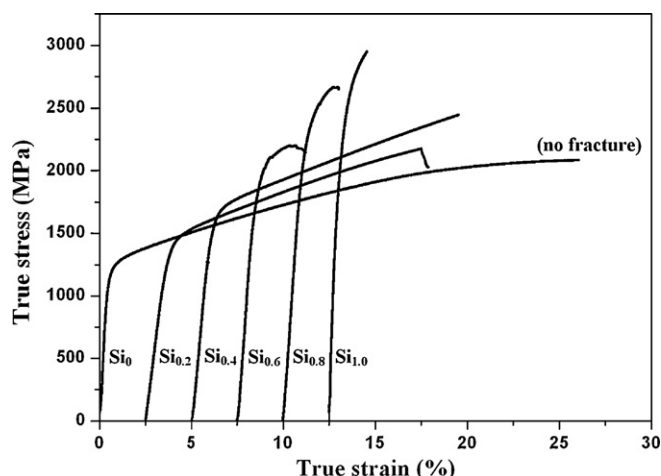


Fig. 4. Compressive true stress-true strain curves of as-cast AlCoCrFeNiSi_x ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1.0) alloys.

Table 3

Lattice constants of BCC phase in as-cast AlCoCrFeNiSi_x ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1.0) alloys.

x	0	0.2	0.4	0.6	0.8	1.0
Lattice constant (Å)	2.878	2.875	2.873	2.866	2.864	2.861

mation of nanoscale cellular structure will result in formation of a great deal of phase interface, which can effectively enhance the alloy strength. According to the Orowan's equation [22], the smaller the spacing of the precipitated phase, the higher the strength, since it hinders effectively the movement of the dislocation. Si₀ alloy exhibits plastic deformation of about 24% without fracture. Si_{0.2} and Si_{0.4} alloy exhibit plastic fracture. The alloy exhibits brittle fracture when Si content is more than 0.4. The precipitated δ phase at grain boundary should be responsible for transition of alloy from plastic fracture to brittle fracture. The fracture mechanism of AlCoCrFeNiSi_x alloy is concluded to be cleavage fracture.

5. Conclusions

In summary, by introducing Si element, AlCoCrFeNiSi_x alloys are designed and studied on its microstructure and mechanical properties. The alloy system is composed mainly of BCC solid solution.

Nanoscale cellular structure is precipitated in the grain of alloys with Si introduction through spinodal decomposition. Si element with small atomic radius exists mainly in solid solution by substituting other elements. Al and Cr elements are mainly substituted by Si element, and segregate in grain and grain boundary. The improvement of mechanical properties is attributed to the solid solution of Si element and precipitation strengthening of nanoscale cellular structure. The alloy transits from plastic fracture to brittle fracture due to the precipitated δ phase at grain boundary. The fracture mechanism of AlCoCrFeNiSi_x alloy is cleavage fracture.

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